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Inventor(s)	Carriere; Marc et al.

Valve, method, and system

Abstract

A valve, including a housing having a port, a valve member positioned to selectively, adjustably, and repeatably obstruct, unobstruct, and partially obstruct the port solely by interaction with an untethered object, an indexer operably connected to the valve member; and an object profile in operable contact with the indexer. A method for managing a wellbore, including dropping an object to land on the object seat of a valve, cycling the indexer, moving the valve member to a position dictated by the indexer, changing the object on the profile, and releasing the object after changing the object. A borehole system, including a plurality of valves, each object profile of the plurality of valves having a different dimension. A wellbore system, including a borehole in a subsurface formation, a string in the borehole, and a valve disposed within or as a part of the string.

Inventors:	Carriere; Marc (Calgary, CA), Hibbs; Justin (Calgary, CA)
Applicant:	Baker Hughes Oilfield Operations LLC (Houston, TX)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9752411	12/2016	Adam	N/A	E21B 34/14
10961818	12/2020	Inglis et al.	N/A	N/A
11454074	12/2021	Kellner et al.	N/A	N/A
2005/0230151	12/2004	McGarian	175/61	E21B 41/0042
2007/0107944	12/2006	Lee	175/317	E21B 10/322
2011/0315389	12/2010	Crider	166/329	E21B 21/103
2012/0227973	12/2011	Hart	166/329	E21B 21/103
2016/0273322	12/2015	Russo	N/A	E21B 43/12
2017/0022783	12/2016	Yong	N/A	E21B 34/10
2023/0279736	12/2022	Eriksen	166/192	E21B 17/1078
2024/0167363	12/2023	Arabskyy	N/A	E21B 34/142

OTHER PUBLICATIONS

AirGlide Floation Sub; 2 pages; Halliburton 2022;
<https://www.halliburton.com/en/products/airglide-floatation-sub>. cited by applicant

Primary Examiner: Bemko; Taras P

Attorney, Agent or Firm: Baker Hughes Company

Background/Summary

BACKGROUND

(1) In the resource recovery and fluid sequestration industries valves are often used to control fluid fronts to optimize production of desired fluids or optimize the formation acceptance of sequestration fluids. Such valves work well for their purposes but require interventions to move them either the first time or after the first time. Interventions generally require a shutdown of operations and the transportation of significant equipment, all of which is time-consuming and expensive. The art would positively receive improvements.

SUMMARY

(2) An embodiment of a valve, including a housing having a port, a valve member positioned to selectively, adjustably, and repeatably obstruct, unobstruct, and partially obstruct the port solely by interaction with an untethered object, an indexer operably connected to the valve member; and an object profile in operable contact with the indexer.

(3) An embodiment of a method for managing a wellbore, including dropping an object to land on the object seat of a valve, cycling the indexer, moving the valve member to a position dictated by

- the indexer, changing the object on the profile, and releasing the object after changing the object.
- (4) An embodiment of a borehole system, including a plurality of valves, each object profile of the plurality of valves having a different dimension.
- (5) An embodiment of a wellbore system, including a borehole in a subsurface formation, a string in the borehole, and a valve disposed within or as a part of the string.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
- (2) FIG. 1 is a side view of a valve as disclosed herein;
- (3) FIG. 2 is the view of FIG. 1 with the outermost portions transparent;
- (4) FIG. 3 is a sectional view illustrating the valve in a closed position;
- (5) FIG. 4 is a sectional view illustrating the valve in an object on seat position with valve components shifted pursuant to hydraulic pressure on the object;
- (6) FIG. 5A is a sectional view of the valve in a fully open position subsequent to object disposal;
- (7) FIG. 5B is a sectional view of the valve in a partially (choked) open position subsequent to object disposal;
- (8) FIG. 6 is an enlarged view of the object seat and object prior to hydraulic pressure application;
- (9) FIG. 7 is the view of FIG. 6 illustrating actuation of an embodiment of an object disposal configuration; and
- (10) FIG. 8 is a view of a borehole system including a valve as disclosed herein.

DETAILED DESCRIPTION

- (11) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.
- (12) Referring to FIGS. 1 and 2, a valve **10** is illustrated in whole (FIG. 1) and with transparent outer components (FIG. 2). The valve **10** includes a housing **12** having one or more ports **14** extending from an inside surface **15** through a wall thickness to an outside surface **16**. An object **18** is illustrated prior to engaging the valve **10**. While the object **18** is illustrated as a ball, this is by no means limiting. The object **18** may be a ball, a dart, or any other untethered object having any geometric shape that might be desired for a particular application. The object **18** must be able to travel through a string (generally illustrated in FIG. 8) connected to the valve **10** to ensure the object **18** can at a selected time engage the valve **10** to have an effect as described hereunder.
- (13) In FIG. 2, the housing **12** has been rendered transparent to illustrate the internal components of the valve **10**. These include a valve member **20** having one or more ports **22** therein. In this embodiment the valve member **20** is illustrated as a sleeve but could be configured differently and have the same effect. The valve member **20** is operably connected to an indexer **24** that may include a J-slot **26** and a biaser **28**, that may be a coil spring or similar. Also included is an object profile **30** that may be configured as an actuating profile or an object seat, for example, that is configured to receive the object **18** and create a pressure drop in fluid applied to the object across the object **18** and seat **30** combined. Such a pressure differential will cause the following components, valve member **20**, indexer **24** and seat **30** to move relative to the housing **12** thereby compressing the biaser **28**. Relief of the applied pressure on the object **18**/object seat **30** will allow the biaser **28** to uncompress and cause an indexing operation of the indexer **24**. It will be appreciated that the path of the J-slot **26** includes camming surfaces **34** that when contacting a cam **36** disposed in the housing **12** causes both rotation of the indexer **24** and longitudinal positioning depending upon how long individual slots of the J-slot **26** are when aligned with the cam **36**. The longitudinal positioning of the indexer **24** causes the same longitudinal positioning of the valve

member **20** (unique to the particular slot). Accordingly, the position of the valve member **20** may be open, closed, choked, and differing levels of choked. The valve member **20** is actuatable in either direction, i.e., toward closed or toward open for the life of the valve **10** and therefore is not limited by movement in a single direction as valves of the art tend to be.

(14) FIGS. 3-5 illustrate a single sequence of movements. FIG. 3 illustrates a run-in closed position, FIG. 4 illustrates landing an object **18** on the seat **30**, where pressure has been applied to compress the biaser **28** and cycle the indexer **24** by one increment or one half increment (depending upon whether the indexer is configured to rotate upon stroke in one direction or is configured to rotate upon stroke in both directions, respectively). FIGS. 5A and 5B illustrate the valve **10** in a new position after disposal of the object **18**, having been through one cycle of movement. FIG. 5A illustrated the valve **10** fully open with ports **14** and **22** all aligned for flow, while FIG. 5B illustrates a choked position where only some of the ports **22** are aligned for flow with ports **14**. It will be appreciated that another object **18** could be landed on the seat **30** to again cycle the valve **10** and obtain a different position of the valve member **20** relative to the housing **12**. The process may be repeated over and over to cycle the valve member position to any of the positions dictated by the indexer (which were decided when the indexer was manufactured).

(15) Disposal of the object **18** has been alluded to above but not described. Any means to remove the object from the seat **30** is contemplated as the ID of the valve **10** needs to be open for flow to occur. While disposal could include flowing the object back to surface (ditto for systems with multiple valves providing the actuation object geometry is unique to each valve **10**), or degrading the object **18** if made at least in part of a degradable material such as a corrodible electrolytic metallic material or degrade-on-demand material both available from Baker Hughes, in accordance with the disclosure hereof, the object **18** may be disposed of by being broken down to a size and geometry that is too small to affect more downhole valves **10** from the one illustrated. In this case, the valve **10** will include an object breaker **40**. In the illustrated embodiment, seen best in FIGS. 6 and 7, where this portion of the valve **10** is enlarged, one embodiment of the breaker **40** includes radially mobile load pins **42** and a geometric profile **44** of the housing **12** that actuates the load pins **42** automatically upon stroking of the seat **30** (compare positions in FIG. 6 and FIG. 7). The load pin **42** applies a point load at point **46** on the object **18** after the object **18**, seat **30**, valve member **20** and indexer **24** are shifted (to the right in the figure (see also FIG. 4). In FIGS. 6 and 7, it will be appreciated that the housing **12** includes the frustoconical inside surface or profile **44** that squeezes the load pins **42** radially inwardly to put the load on the object **18**. The point loads on the object **18** cause fracturing of the object **18** and disassociation of the object **18** into a plurality (or a multiplicity) of smaller pieces. In embodiments, the object may employ a frangible material, as a material for a portion or the whole of the object **18**.

(16) The material may be glass, ceramic, carbide, porcelain, alloy, composites, plastic, metallic, polymer, cement, etc. providing the materials are frangible. "Frangible" for purposes hereof means the material will rupture or crumble into smaller pieces when impacted or sufficiently loaded against a stress riser. Combinations of materials including at least one of the materials hereinbefore identified are also contemplated. In embodiments, the load pins **42** are positioned to ensure the resulting pieces are too small in overall dimensions to actuate more downhole located valves **10**. Load pins **42** may number one to as many as can be physically disposed in the seat **30**. In FIG. 2, it is apparent that 4 are illustrated, for example. Configured as disclosed, the valve **10** is cyclable one time per object dropped with positioning of the valve subsequent to each cycle being dictated only by the J-slot **26**. Additional valves **10** disposed further downhole from the one illustrated (but appear substantially as illustrated for the one in the figures) will employ a different geometry object **18** such as, for example, a smaller diameter ball. In each case, the object will be disposed of after a single pressure event to cycle that specific valve **10** in which the object was able, by geometry, to seat.

(17) Referring to FIG. 8, a borehole system **60** is illustrated. The system **60** comprises a borehole

62 in a subsurface formation **64**. A string **66** is disposed within the borehole **62**. One or more valves **10** as disclosed herein is/are disposed within or as a part of the string **66**.

(18) Set forth below are some embodiments of the foregoing disclosure:

(19) Embodiment 1: A valve, including a housing having a port, a valve member positioned to selectively, adjustably, and repeatably obstruct, unobstruct, and partially obstruct the port solely by interaction with an untethered object, an indexer operably connected to the valve member; and an object profile in operable contact with the indexer.

(20) Embodiment 2: The valve as in any prior embodiment, wherein the valve member is a sleeve,

(21) Embodiment 3: The valve as in any prior embodiment, wherein the indexer includes a J-slot.

(22) Embodiment 4: The valve as in any prior embodiment, wherein the indexer includes a biaser.

(23) Embodiment 5: The valve as in any prior embodiment, wherein the valve further includes an object breaker.

(24) Embodiment 6: The valve as in any prior embodiment, wherein the object breaker is adjacent the object profile.

(25) Embodiment 7: The valve as in any prior embodiment, wherein the object breaker includes a load pin that is loaded against an object landed on the object profile when the object profile is displaced in the housing.

(26) Embodiment 8: The valve as in any prior embodiment, wherein the load pin is radially movable.

(27) Embodiment 9: The valve as in any prior embodiment, wherein the breaker includes a plurality of load pins.

(28) Embodiment 10: The valve as in any prior embodiment, wherein the plurality of load pins are arranged circumferentially about the object profile.

(29) Embodiment 11: The valve as in any prior embodiment, wherein the untethered object includes a frangible material.

(30) Embodiment 12: A method for managing a wellbore, including dropping an object to land on the object seat of a valve as in any prior embodiment, cycling the indexer, moving the valve member to a position dictated by the indexer, changing the object on the profile, and releasing the object after changing the object.

(31) Embodiment 13: The method as in any prior embodiment, wherein the changing is breaking.

(32) Embodiment 14: The method as in any prior embodiment, wherein the breaking is into a multiplicity of pieces.

(33) Embodiment 15: The method as in any prior embodiment, wherein the object includes a frangible material.

(34) Embodiment 16: The method as in any prior embodiment, wherein the changing is in a way that renders the object incapable of landing on a profile of another tool downhole of the valve.

(35) Embodiment 17: The method as in any prior embodiment, wherein the another tool is another valve.

(36) Embodiment 18: The method as in any prior embodiment, wherein the moving includes to a position among fully open flow, choked flow, or closed to flow and from any of fully open flow, choked flow, or closed to flow to another of fully open flow, choked flow, or closed to flow according to a each cycle of the indexer.

(37) Embodiment 19: A borehole system, including a plurality of valves as in any prior embodiment, each object profile of the plurality of valves having a different dimension.

(38) Embodiment 20: A wellbore system, including a borehole in a subsurface formation, a string in the borehole, and a valve as in any prior embodiment disposed within or as a part of the string.

(39) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Further, it should be noted that the terms “first,” “second,” and the like herein do not denote any

order, quantity, or importance, but rather are used to distinguish one element from another. The terms “about”, “substantially” and “generally” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” and/or “substantially” and/or “generally” can include a range of $\pm 8\%$ of a given value.

(40) The teachings of the present disclosure may be used in a variety of well operations. These operations may involve using one or more treatment agents to treat a formation, the fluids resident in a formation, a borehole, and/or equipment in the borehole, such as production tubing. The treatment agents may be in the form of liquids, gases, solids, semi-solids, and mixtures thereof. Illustrative treatment agents include, but are not limited to, fracturing fluids, acids, steam, water, brine, anti-corrosion agents, cement, permeability modifiers, drilling muds, emulsifiers, demulsifiers, tracers, flow improvers, Non compressible gases, etc. Illustrative well operations include, but are not limited to, hydraulic fracturing, stimulation, tracer injection, cleaning, acidizing, steam injection, water flooding, cementing, etc.

(41) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

Claims

1. A valve, comprising: a housing having a housing port; a valve member having a valve port, the valve member positioned to selectively, adjustably, and repeatably, in fixed positions, obstruct by misaligning the valve port with the housing port, unobstruct by fully aligning the valve port with the housing port, and partially obstruct by partially aligning the valve port with the housing port solely by interaction with a sequentially dropped one of a plurality of untethered objects having a substantially same size; an indexer operably connected to the valve member; and an object profile in operable contact with the indexer, the profile configured to impart a mechanical load on the object to fragment the object while on the profile.
2. The valve as claimed in claim 1, wherein the valve member is a sleeve.
3. The valve as claimed in claim 1, wherein the indexer includes a J-slot.
4. The valve as claimed in claim 1, wherein the indexer includes a biaser.
5. The valve as claimed in claim 1, wherein the valve further includes an object breaker.
6. The valve as claimed in claim 5, wherein the object breaker is adjacent the object profile.
7. The valve as claimed in claim 5, wherein the object breaker includes a load pin that is loaded against an object landed on the object profile when the object profile is displaced in the housing.
8. The valve as claimed in claim 7, wherein the load pin is radially movable.
9. The valve as claimed in claim 7, wherein the breaker includes a plurality of load pins.
10. The valve as claimed in claim 9, wherein the plurality of load pins are arranged circumferentially about the object profile.
11. The valve as claimed in claim 1, wherein the untethered object includes a frangible material.
12. A method for managing a wellbore using an object of a plurality of objects having a substantially same size, comprising: dropping the object to land on the object seat of a valve as

claimed in claim 1; cycling the indexer; moving the valve member to a position dictated by the indexer; imparting a mechanical load on the object to fragment the object while on the profile; and releasing the object through the unchanged profile after fragmenting the object.

13. The method as claimed in claim 12, wherein the changing is breaking.

14. The method as claimed in claim 13, wherein the breaking is into a multiplicity of pieces.

15. The method as claimed in claim 12, wherein the object includes a frangible material.

16. The method as claimed in claim 12, wherein the changing is in a way that renders the object incapable of landing on a profile of another tool downhole of the valve.

17. The method as claimed in claim 16, wherein the another tool is another valve.

18. The method as claimed in claim 12, wherein the moving includes to a position among fully open flow, choked flow, or closed to flow and from any of fully open flow, choked flow, or closed to flow to another of fully open flow, choked flow, or closed to flow according to a each cycle of the indexer.

19. A borehole system, comprising: a plurality of valves as claimed in claim 1, each object profile of the plurality of valves having a different dimension.

20. A wellbore system, comprising: a borehole in a subsurface formation; a string in the borehole; and a valve as claimed in claim 1 disposed within or as a part of the string.
